

Advanced Lesson 3 - Exercises: Solve, Plot, Differentiate, Integrate, Simplify Mikey Ferguson MATH 7013 - Advanced Mathematical Modeling

Everyday notation has been converted to Mathematica notation throughout: equations use "==", functions use square brackets, user definitions use the underscore, and E, Pi and I are the built-in constants rather than the bare letters e, p and i.

```
Clear[a, b, c, x, y, f, n];
```

1: Solving a biquadratic

```
Solve[a x^4 + b x^2 + c == 0, x]
```

```
Solve[a x^4 + b x^2 + c == 0, a]
```

2: A nonlinear system in two unknowns

```
sol2a = Solve[{a x^2 + b y + c == z, y + a z^2 + b x^2 + c == 1}, {x, y}]
```

```
Simplify[sol2a]
```

```
sol2b = Solve[{a x^2 + b y + c == z, y + a z^2 + b x^2 + c == 1}, {a, x}]
```

3: Plot and integrate $E^{(2x)} \sin[3x] \cos[4x]$

```
Plot[E^2 x Sin[3 x] Cos[4 x], {x, 0, 2}]
```

```
int3 =  $\int e^{2x} \sin[3x] \cos[4x] dx$ 
```

```
Simplify[int3]
```

4: The same function, ten ways

```
Plot[E^2 x Sin[3 x] Cos[4 x], {x, 0, 2}]
```

```
d4 = D[E^2 x Sin[3 x] Cos[4 x], x]
```

```
Simplify[d4]
```

```
Clear[f, x];
```

```
f[x_] = E^2 x Sin[3 x] Cos[4 x];
```

```
Plot[f[x], {x, 0, 2}]
```

```
f'[x]
```

```
Simplify[f'[x]]
```

```
{Simplify[f'[x] - D[f[x], x]}
```

```
{Simplify[f''[x] - D[f'[x], x]}
```

```
{Simplify[f'''[x] - D[f''[x], x]}
```

5: The local maximum of $x^3 - x + 1$

```
Clear[f, x];
```

```
f[x_] = x^3 - x + 1;
```

```
{f'[x], D[f[x], x], Simplify[f'[x] - D[f[x], x]}
```

```
{f''[x], D[f'[x], x], Simplify[f''[x] - D[f'[x], x]}
```

```
Plot[{f[x], f'[x]}, {x, -2, 2}, PlotLegends -> {"f", "f'"}]
```

```
crit = Solve[f'[x] == 0, x]
```

```
Table[{x /. s, f''[x] /. s, f[x] /. s}, {s, crit}]
```

6: Simplifying $\cos[n\pi]$ under assumptions

```
Clear[n];
```

```
Simplify[Cos[n Pi]]
```

```
Simplify[Cos[n Pi], n ∈ Z]
```

```
Simplify[Cos[n Pi], n/2 ∈ Z]
```

```
Simplify[Cos[n Pi], (n-1)/2 ∈ Z]
```

7: Plot and integrate $x^5 \sin[x]$

```
Plot[x^5 Sin[x], {x, -1, 1}]
```

$$\int x^5 \sin[x] \, dx$$

$$\int_2^3 x^5 \sin[x] \, dx$$

8: Sqrt[x] and the trouble with negative limits

```
Plot[Sqrt[x], {x, -1, 1}, PlotStyle -> Thickness[0.01`]]
```

$$\int \sqrt{x} \, dx$$

$$\int_1^2 \sqrt{x} \, dx$$

$$\int_a^b \sqrt{x} \, dx$$

```
Integrate[Sqrt[x], {x, a, b}, Assumptions -> 0 < a < b]
```

```
Integrate[Sqrt[x], {x, a, b}, Assumptions -> a < 0 < b]
```